My wireless motion sensor solution

The solution consists of the following two hardware components:





- SONOFF ZBDongle-P Zigbee Coordinator (the central hub that manages the Zigbee network)
- Shelly BLU Motion ZB Sensor (sends motion data to the SONOFF ZBDongle-P through Zigbee, which is then published as MQTT messages)
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I chose Zigbee2MQTT as the wireless communication solution for this project.

How It Works

Zigbee2MQTT acts as a translator between Zigbee devices and Python applications.

- The Shelly BLU Motion ZB communicates using the Zigbee protocol.
- Python applications communicate through TCP/IP, MQTT (for example, Mosquitto), files, databases, APIs, and other protocols. In my case it is MQTT.
- The SONOFF ZBDongle-P functions as the Zigbee radio coordinator.
- Zigbee2MQTT converts Zigbee messages into MQTT messages that Python applications can easily read and process.

Zigbee is a low-power, short-range wireless communication standard designed for smart home and Internet of Things (IoT) devices.

Zigbee Overview

Mesh Network

Unlike Wi-Fi, where each device connects directly to a central router. Zigbee creates a self-healing mesh network (in this case SONOFF ZBDongle-P Zigbee Coordinator).

Device Roles

Coordinator

- The central hub that creates and manages the Zigbee network (SONOFF ZBDongle-P Zigbee Coordinator).

Routers

- Mains-powered devices, such as smart plugs and light bulbs.
- Receive and forward data from other Zigbee devices.
- Extend the network's range.

End Devices

- Battery-powered devices, such as motion sensors and door sensors.
- Spend most of their time in sleep mode to conserve power.

Low Power Consumption
Zigbee operates primarily on the 2.4 GHz frequency band (along with regional 868 MHz and 915 MHz bands).
The maximum data rate is 250 kbps, allowing battery-powered sensors to operate for years on a single battery.

Security
Zigbee uses 128-bit AES encryption to secure communications between devices.

Installation

1. Connect the Zigbee Coordinator

Plug the SONOFF ZBDongle-P Zigbee Coordinator into the Raspberry Pi 5.
Determine the device´s serial device path:
- ls -l /dev/serial/by-id/

2. Install Mosquitto MQTT Broker

Install Mosquitto and its client tools:
- sudo apt install mosquitto mosquitto-clients -y
Enable and start the service:
- sudo systemctl enable mosquitto
- sudo systemctl start mosquitto

Verify that the service is running:
- sudo systemctl status mosquitto

3. Install Zigbee2MQTT

Create a working directory:
-mkdir ~/zigbee2mqtt
- cd ~/zigbee2mqtt

Install the latest LTS version of Node.js:
- curl -fsSL https://deb.nodesource.com/setup_lts.x | sudo bash -
This command installs the NodeSource repository for the latest Long-Term Support (LTS) version of Node.js.

Install Node.js:
- sudo apt install nodejs -y

Clone the Zigbee2MQTT repository:
- git clone https://github.com/Koenkk/zigbee2mqtt.git
Zigbee2MQTT provides a web-based user interface for managing Zigbee devices and monitoring the Zigbee network.

Navigate to the data directory:
- cd zigbee2mqtt/data

Edit the configuration file:
- cp configuration.example.yaml configuration.yaml
- nano configuration.yaml
Configure your MQTT server and Zigbee coordinator settings, check my configuration.yaml for reference under the Zigbee2MQTT folder.

Install dependencies:
- pnpm install
- pnpm run build
Start Zigbee2MQTT:
- npm start

Open the Zigbee2MQTT web interface:
http://raspberry_pi_ip:8080, For more information, see, https://github.com/Koenkk/zigbee2mqtt

Configure Zigbee2MQTT to start automatically
Create a systemd service file:
- sudo nano /etc/systemd/system/zigbee2mqtt.service

Service file's contents:
[Unit]
Description=Zigbee2MQTT
After=network.target

[Service]
Type=simple
User=camerauser
WorkingDirectory=/home/camerauser/zigbee2mqtt/zigbee2mqtt
ExecStart=/usr/bin/npm start
Restart=always

[Install]
WantedBy=multi-user.target

Create the service.
- sudo systemctl daemon-reload
- sudo systemctl enable zigbee2mqtt.service
- sudo systemctl start zigbee2mqtt.service
Check the service status:
- sudo systemctl status zigbee2mqtt.service

Node.js
Node.js is a cross-platform, open-source JavaScript runtime environment that runs on Windows, Linux, Unix, macOS, and other operating systems.
It uses Google's V8 JavaScript engine to execute JavaScript code outside of a web browser.

pnpm

pnpm is a package manager for Node.js.

The name originally stood for "Performant npm", reflecting its goal of being a faster and more efficient alternative to npm.

Compared to npm, pnpm:

- Installs packages faster
- Uses less disk space
- Handles dependencies more efficiently
- Is used by modern projects such as Zigbee2MQTT